SKILLS GAP ANALYSIS

Preparation for Entry-Level SOC and Cybersecurity Roles

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COSC 4272 — Senior Seminar
Research date: November 13, 2026

1. Current Career Objective

My immediate goal is to begin in an entry-level SOC Analyst or IT security role after completing my Computer Science degree at Lamar University, expected December 2026. I am particularly interested in alert analysis, incident response, security monitoring, and digital forensics. The job postings reviewed for this project show that employers value networking and operating-system fundamentals, SIEM and EDR tools, incident documentation, security certifications, and clear communication.

2. Current Strengths

• **Programming and scripting foundation:** Coursework and projects using Java, C++, C, and Python; experience reading and modifying code and building a Python/Streamlit AI cybersecurity assistant.

• **Operating systems and virtualization:** Linux/Ubuntu and Windows familiarity, with VirtualBox experience from academic lab environments.

• **Networking fundamentals:** Coursework and hands-on work involving TCP/IP, client-server sockets, IP addressing, ARP, and network troubleshooting.

• **Cybersecurity project experience:** CSRF attack simulation and token-protection lab, PIN Security Simulator, and CyberGuard AI alert-analysis project.

• **Databases and development tools:** MySQL, Node.js/Express, Postman, Git, GitHub, and Eclipse experience through coursework and projects.

• **Teamwork and communication:** Academic team projects, technical documentation, project coordination, and professional internship exposure to digital-forensics topics.

3. Skills Gaps Compared with Target Roles

Skill / priority	Gap / evidence	Development action
SIEM tools and log queries Priority: High	The postings mention SIEM platforms such as Splunk, Microsoft Sentinel, QRadar, Sumo Logic, and query languages. I do not yet have sustained hands-on experience with a production SIEM.	Complete guided labs in one SIEM first; practice searches, alert triage, dashboards, and concise investigation notes.
Endpoint detection and response (EDR) Priority: High	Binary Defense asks for endpoint-protection platform familiarity. I need practical exposure to endpoint alerts and investigation workflows.	Study endpoint telemetry and complete a safe lab or vendor training module using a trial/community environment where available.
Incident response workflow Priority: High	SOC roles require alert validation, severity assessment, evidence collection, escalation, containment coordination, and handoff documentation.	Practice a repeatable alert-triage workflow and write short incident reports for phishing, brute-force, and suspicious-login scenarios.
Security certification Priority: High	Some postings prefer or require Security+, CySA+, GSEC, or another cybersecurity certification. I am beginning Security+ preparation and do not currently hold a reported certification.	Build Security+ foundations with a structured study schedule; schedule an exam only after practice scores and readiness support it.
Cloud security fundamentals Priority: Medium	Cloud exposure (AWS, Azure, and GCP) appears in some remote SOC listings and is useful for modern security monitoring.	Learn cloud identity, logging, network controls, and basic security monitoring in one platform before expanding to others.
OT/ICS and SCADA Priority: Medium for Jacobs; lower for general SOC	The Jacobs role emphasizes industrial systems, SCADA, OT networks, and standards such as IEC 62443 and NERC CIP.	Study OT/ICS architecture, the Purdue model, common industrial protocols, and the purpose of relevant standards.
Advanced security analytics and query languages Priority: Medium	Roles require investigating events efficiently using search/query syntax and recognizing patterns in logs.	Practice foundational SPL or KQL queries, filtering, time ranges, aggregation, and correlation in a lab.
Professional SOC experience Priority: High	Some roles ask for 0–3 years of SOC experience or at least one year of security operations or equivalent experience.	Use labs, documented projects, internships, and volunteer/academic security work to demonstrate applied skills without presenting them as paid SOC employment.

4. Development Plan: 12 Weeks

Available study time: approximately 2–3 hours per week. The plan emphasizes practical evidence that can be shown in a portfolio.

Weeks 1–4: Foundations

• Study Security+ domains for approximately 45–60 minutes each week.

• Complete introductory SIEM/log-analysis labs for approximately 60 minutes each week.

• Write two short investigation notes describing the alert, evidence reviewed, likely severity, and recommended next step.

Weeks 5–8: Applied SOC practice

• Continue Security+ preparation and review missed practice questions.

• Practice basic SPL or KQL searches and examine Windows/Linux log examples.

• Complete one phishing or suspicious-login investigation and one endpoint-alert exercise; document findings and escalation decisions.

Weeks 9–12: Specialization and portfolio

• Learn introductory cloud security logging in either Azure or AWS.

• For OT/ICS opportunities, study the Purdue model, SCADA components, and introductory IEC 62443/NERC CIP concepts.

• Publish sanitized lab write-ups and investigation reports to GitHub; update the resume with accurate project outcomes.

• Review job postings again and apply to roles whose required qualifications are reasonably met.

5. Measurable Outcomes

• Complete at least four guided SIEM/log-analysis exercises and document the searches used.

• Produce at least three concise incident-investigation write-ups with evidence, assessment, and recommended response.

• Complete a focused Security+ study cycle and track practice-test performance; make an exam decision based on readiness.

• Add at least two polished, sanitized security-lab write-ups to the GitHub portfolio.

• Demonstrate foundational familiarity with one cloud platform and, if pursuing OT security, introductory OT/ICS terminology.

6. Conclusion

My foundation in programming, Linux, networking, academic cybersecurity labs, and project development provides a useful starting point for entry-level security work. The most important gaps are practical SIEM and EDR experience, structured incident-response practice, and an industry certification. A focused 12-week plan of 2–3 hours per week can help convert coursework and projects into stronger evidence of job readiness. I will represent my experience accurately and prioritize roles that value academic projects, internships, and early-career technical skills.

Sources Consulted

• Cyber Security Analyst I — CyberSheath: https://job-boards.greenhouse.io/cybersheath/jobs/5473521008

• Associate SOC Analyst — Expel: https://expel.com/about/careers/

• Tier 1 SOC Analyst — Binary Defense: https://www.indeed.com/viewjob?jk=68c7ff4c5af7bb5b

• Jr. OT Cybersecurity Engineer — Jacobs: https://www.indeed.com/viewjob?jk=39db8a4b7465bdb8

• Enterprise IT Security Analyst I — University of Houston System: https://careers.uh.edu/jobs/search/uh-central-staff

Job-market review date: November 13, 2026.